ME5413 Report

Homework 1: Perception

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I. Introduction

Perception is a critical part in robot system design and implementation. With the development of Robotics technology, the boundaries of robot perception have been greatly expanded. Given serval data to process, this assignment consists of 3 tasks, each of the task may or may not consist of some subtasks, which are as follows:

- Task1: Single Object Tracking (SOT)
  - Task1.1: SOT based on Template Matching
  - Task1.2: SOT based on Object Detection with Association
  - Task1.3: SOT based on YOLOv5-DeepSORT
- Task2: Multi Object Prediction
  - Task2.1: Prediction based on Constant Velocity Model
  - Task2.2: Prediction based on Constant Acceleration Model
- Task3: Single Object Tracking in ROS

In task1, we successfully implement 3 SOT methods with comparison, where YOLOv5-DeepSORT obtains the best average Precision score 0.7591 and average Success score 0.9033. In task2, we filter the given data, successfully implement CVM and CAM based multi object trajectory prediction, and visualize the prediction results compared with the ground truth for all scenarios. Finally we achieve relatively well performance with both CVM and CAM. In task3 we successfully publish the tracking bounding box and the ground truth information by reading ROS bag files, together with the revised image where these two bounding boxes attached to the original image. Please view the README.md file and check all the experimental results.

II. Methodology

A. *CSRT Tracker*

In task1.1, we utilize the Discriminative Correlation Filter with Channel and Spatial Reliability Tracker (CSRT) to achieve more accurately single object tracking. CSRT is an advanced template matching-based single object tracking algorithm. It extends traditional Discriminative Correlation Filter (DCF) techniques, by introducing spatial reliability maps, which weigh different regions of the target object to reduce boundary effects and improve tracking accuracy. Its good feature is its higher accuracy compared to conventional DCF-based trackers, particularly in scenarios involving occlusions, scale variations, and background clutter. However, one of its main disadvantages is the computational complexity, leading to relatively slow tracking outcomes.

B. *DeepSORT-based ODA methods*

To do single object tracking, the process of models of Object Detection with Association (ODA) can be divided into two stages: (1) object detection, to firstly detect an object in each frame, and (2) association, to associate the detect result (usually a bounding box) across frames. In task1.2 we utilize a pre-trained DETR model[1] and in task1.3 we utilize a pre-train YOLOv5[2] model to do object detection respectively.

Then to do association, Deep Simple Online and Realtime Tracker (DeepSORT) is used in both task1.2 and task1.3. DeepSORT is one of the most well-known methods in the association stage of ODA. It enhances the original Simple Online and Realtime Tracker (SORT), which relies purely on the Kalman filter and Hungarian algorithm, by incorporating deep learning-based appearance features.

C. *CVM and CAM*

In task2.1 and task2.2, Constant Velocity Model (CVM) and Constant Acceleration Model (CAM) are used respectively to predict the multi-agent trajectories. Both of them are physics-based object trajectory prediction methods and their formulas in 2-dimension space are shown as follows:

- CVM:

$$x = v_{x0}t + x_0,$$

$$y = v_{y0} + y_0. \tag{1}$$

- CAM:

$$x = \frac{1}{2} a_{x0} t^2 + v_{x0} t + x_0,$$
$$y = \frac{1}{2} a_{y0} t^2 + v_{y0} t + y_0. \tag{2}$$

III. Experiments and Results

A. *Task1: Single Object Tracking*

The data for Task 1 consists of 5 image sequences and the initial positions of the corresponding targets to be tracked (in the format of x, y, width, height). Each image sequence contains 150 images in jpg format. To evaluate the tracking effect, One-Pass Evaluation Metric's Precision and Success metrics are used. The results with comparison between Template Matching, Object Detection with Association and YOLOv5-DeepSORT can be seen as follows.

Table 1 Evaluation of SOT methods using Precision metrics

		Seq1	Seq2	Seq3	Seq4	Seq5	Average Score
1	Template Matching	1.0	0.8993	0.4899	0.0604	0.7315	0.6362
2	Object Detection with Association	0.9712	1.0	0.3358	0.8714	0.3172	0.6991
3	YOLOv5-DeepSORT	1.0	0.9913	0.3387	0.9231	0.5423	0.7591

Table 2 Evaluation of SOT methods using Success metrics

		Seq1	Seq2	Seq3	Seq4	Seq5	Average Score
1	Template Matching	1.0	0.8322	1.0	0.0604	0.7383	0.7262
2	Object Detection with Association	0.9712	0.9091	0.6194	0.9357	0.9379	0.8747
3	YOLOv5-DeepSORT	1.0	0.9130	0.6694	0.9692	0.9648	0.9033

B. *Task2: Multi Object Prediction*

1) *Data Analysis and Filtering*

In task2, the data corresponding to 287 scenes stores several agents whose trajectories are to be predicted. Each agent provides a total of 9.1 seconds of ground truth trajectories with an interval of 0.1s. However, some of these data are invalid (usually represented by the 0 value that is separated from the context data). If they are retained, the subsequent predicted trajectory will deviate greatly from the valid ground truth trajectory. So a necessary part of this task is to identify this kind of invalid data according to a verification bit not only at the prediction stage but also the evaluation stage, and that eliminate it. Since the predictions of both CVM and CAM are crucially dependent on the initial state of an agent at a specific time point, the initial state of an agent is strictly checked in this task. If it is judged invalid, it will be judged again at a later time point until valid data is obtained as the initial state.

2) Task2.1: Prediction based on Constant Velocity Model

The Average Displacement Error (ADE) and Final Displacement Error (FDE) of CVM prediction results averaged across all target agents and scenarios are shown as follows.

Table 3 Average ADE and FDE across all target agents and scenarios using CVM

	3s	5s	8s
ADE	1.4032	3.6907	7.8303
FDE	3.8956	10.6748	23.0422

3) Task2.2: Prediction based on Constant Acceleration Model

The ADE and FDE of CVM prediction results averaged across all target agents and scenarios are shown as follows.

Table 4 Average ADE and FDE across all target agents and scenarios using CAM

	3s	5s	8s
ADE	4.2483	11.8750	26.7682
FDE	12.4997	35.5642	83.9731

C. Task3: Single Object Tracking in ROS

In task3 we create a ROS space and write a Python script to read the ROS bag files and then publish 4 ROS topic according to the assignment requirement. The results can be seen using the ROS bag files in task1_tracking/results/4_rosbags.

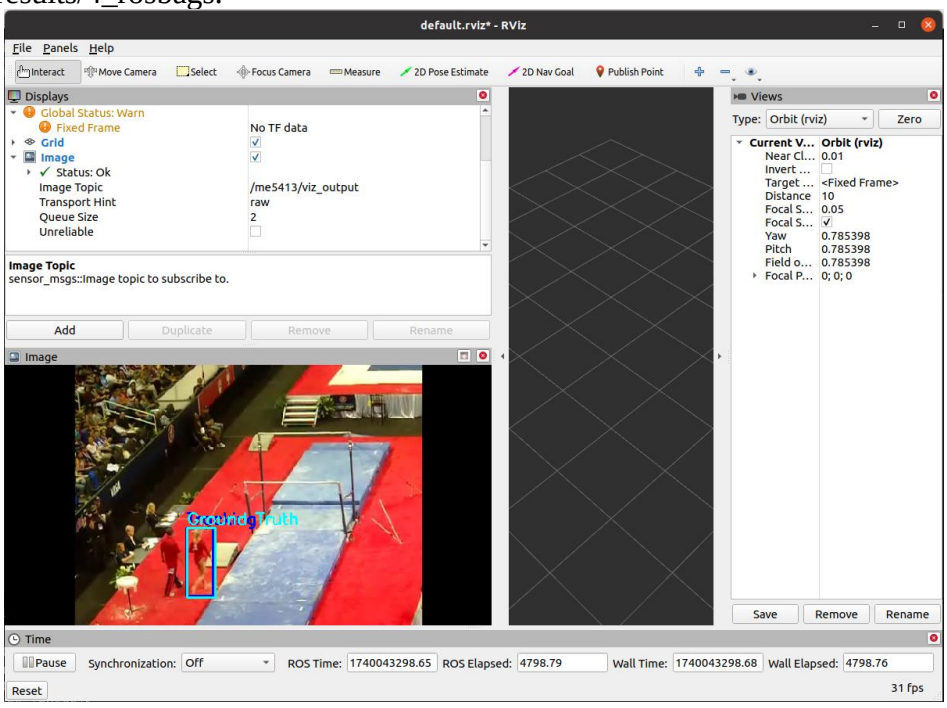

Figure 1 Rviz visualization of SOT in ROS with seq2

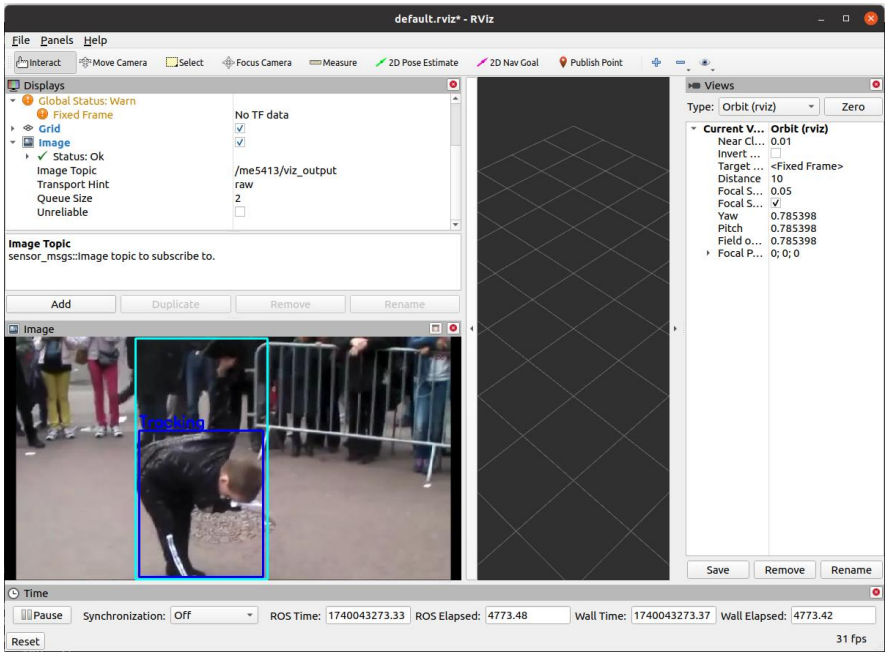

Figure 2 Rviz visualization of SOT in ROS with seq3

IV. Discussion

In task1 we utilize Template Matching, Object Detection with Association(ODA) and YOLOv5-DeepSORT methods, where YOLOv5-DeepSORT obtains the best average Precision score 0.7591 and average Success score 0.9033, indicating superior tracking performance in terms of maintaining accurate object localization. ODA achieves an average precision of 0.6991, performing slightly better than Template Matching (0.6362). Note that the Precision score and Success score of Template Matching at seq4 have a very noticeable sharp drop. We observe that this is because the target volleyball player in the beginning of the image sequence walked past another volleyball player with an occlusion, and the tracking box then marked the wrong player. Whereas each detected objects are learnt by a neural network and marked by an unique number in ODA and YOLOv5-DeepSORT using DeepSORT, tracked objects are less affected by occlusion and deformation.

In task2, we filter the given data and achieve relatively well performance with both CVM and CAM. CVM shows relatively lower ADE and FDE values across all time horizons. The increasing gap between the two models as time progresses suggests that CAM-based predictions may be too simplistic for accurate long-term predictions.

In task3 we read the information from ROS bag files, and successfully publish the tracking bounding box information and the ground truth information, together with the revised image where these two bounding boxes attached to the original image.

V. Conclusion

In this assignment we successfully implement several perception algorithms as well as incorporate them with realistic data and ROS, visualize the results and evaluate them with comparison.

Reference

[1] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, “End-to-End Object Detection with Transformers,” May 28, 2020, *arXiv*: arXiv:2005.12872. doi: 10.48550/arXiv.2005.12872.

[2] “ultralytics/yolov5: YOLOv5 🚀 in PyTorch > ONNX > CoreML > TFLite.” Accessed: Feb. 20, 2025. [Online]. Available: https://github.com/ultralytics/yolov5